EXPERIMENT NO: 02

Upload and Analyze the data set given in CSV Format and Perform Data Preprocessing and Visualization

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#Load the data into a pandas DataFrame
file_path='Sales_data.csv'
df=pd.read_csv(file_path)
#Display the first few rows of the Dataframe
print(df.head())
#Check for missing values
print(df.isnull().sum())
#Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(),inplace=True)
df.dropna(subset=['Product','Quantity','Region'],inplace=True)
#Summary Statistics
print(df.describe())
#Group by product and calculate the total sales and quantity
product_summary=df.groupby('Product').agg({
    'Sales':'sum',
    'Quantity':'sum'
}).reset_index()
print(product_summary)
#Bar plot of total sales by product
plt.figure(figsize=(10,6))
plt.bar(product_summary['Product'],product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

#line plot of sales over time
df['Date']=pd.to_datetime(df['Date'])
sales_over_time=df.groupby('Date').agg({
    'Sales':'sum'
```

```
}).reset_index()
plt.figure(figsize=(10,6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

#Pivot table to analyze sales by region and product
pivot_table=df.pivot_table(values='Sales',index='Region',columns='Product'
,aggfunc=np.sum,fill_value=0)
print(pivot_table)
#correlation matrix
correlation_matrix=df.corr(numeric_only=True)
print(correlation_matrix)

#heatmap of the correlation matrix
plt.figure(figsize=(8,6))
sns.heatmap(correlation_matrix,annot=True,cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

Output:

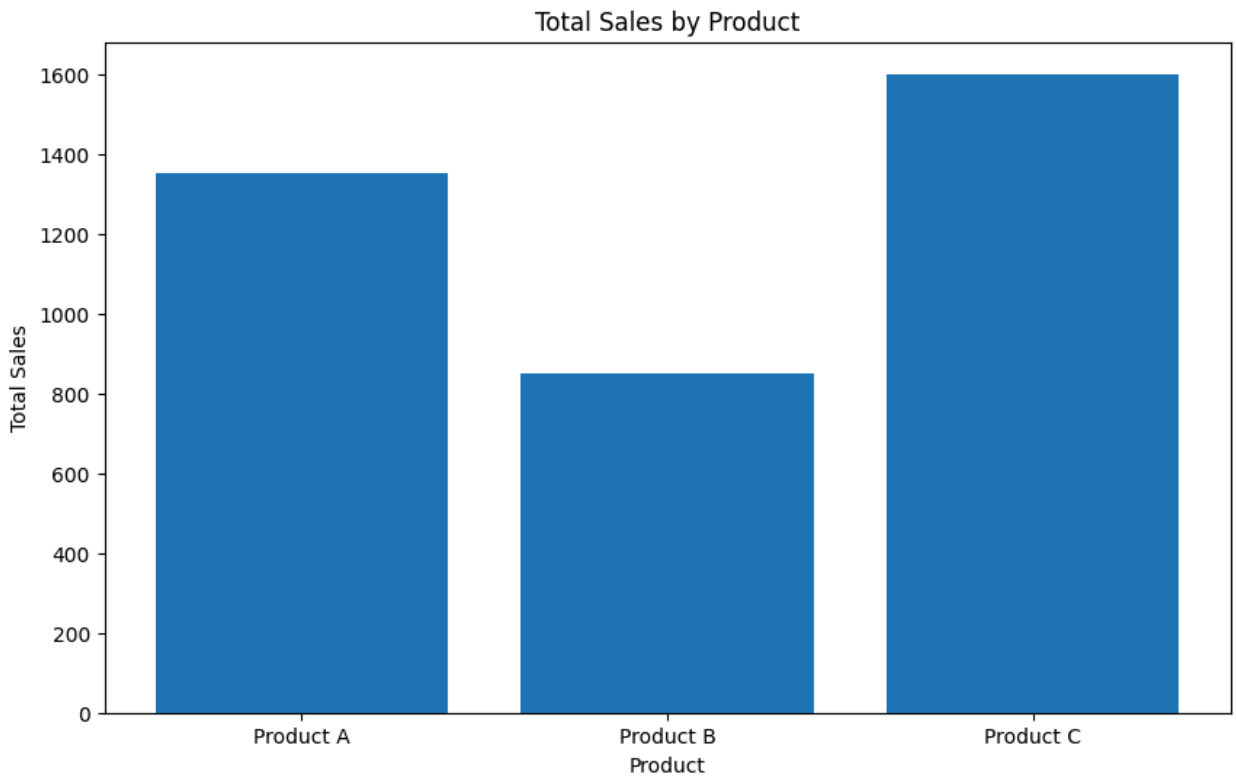
```
        Date    Product  Sales  Quantity Region
0 2023-01-01  Product A    200         4  North
1 2023-01-02  Product B    150         3  South
2 2023-01-03  Product A    220         5  North
3 2023-01-04  Product C    300         6   East
4 2023-01-05  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales   Quantity
count   16.000000  16.000000
```

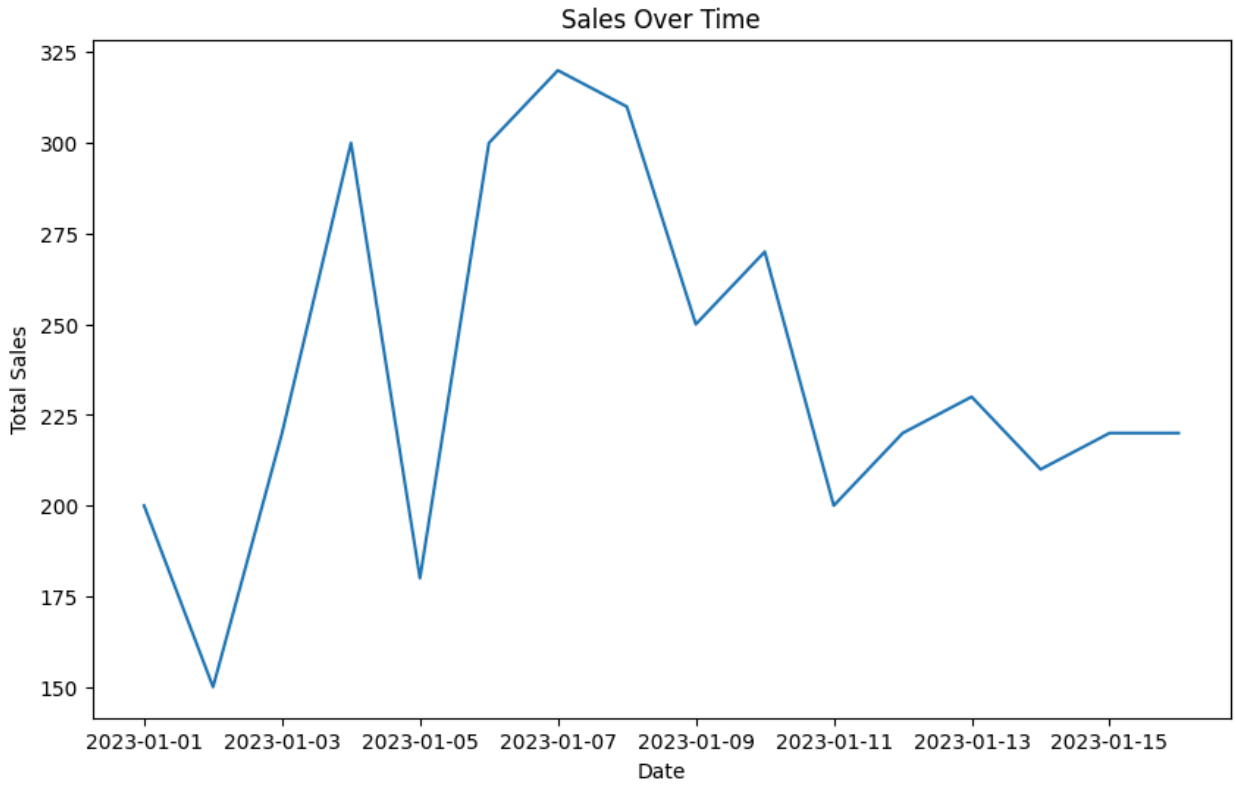
```
mean   237.500000   5.375000
std     49.598387   1.454877
min    150.000000   3.000000
25%    207.500000   5.000000
50%    220.000000   5.000000
75%    277.500000   5.250000
max    320.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```

```
/tmp/ipykernel_3443/1309037498.py:13: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.
```

```
df['Sales'].fillna(df['Sales'].mean(),inplace=True)
```



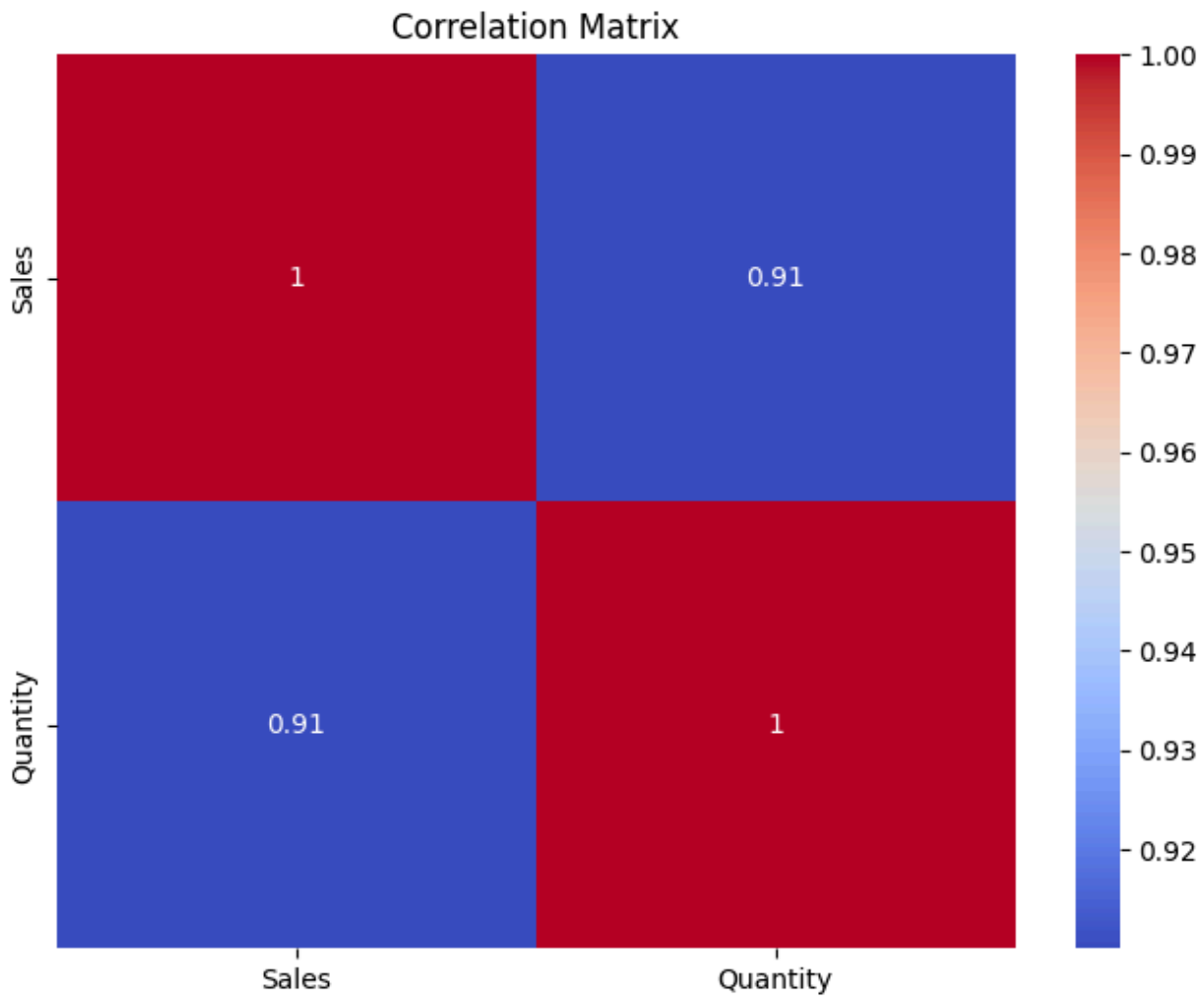


/tmp/ipykernel_3443/1309037498.py:44: FutureWarning: The provided callable <function sum at 0x7f764e80ab60> is currently using

DataFrameGroupBy.sum. In a future version of pandas, the provided callable will be used directly. To keep current behavior pass the string "sum" instead.

```
pivot_table=df.pivot_table(values='Sales',index='Region',columns='Product'
,aggfunc=np.sum,fill_value=0)
```

```
Product  Product A  Product B  Product C
Region
East           310          0        740
North          720          0        410
South          320        420        220
West             0        430        230
            Sales  Quantity
Sales     1.000000  0.910021
Quantity  0.910021  1.000000
```


Correlation Matrix
Sales
Quantity
1
0.91
0.91
1
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92